

Deep learning: Portfolio assignment 3

Jeffrey Gerrits - Master of Applied Data Science UOS3 - 07-03-2026

In this experiment, 16 model configurations were tested to improve a recurrent neural network (RNN). The goal was to understand the effects of different configurations.

The first results looked almost too good. After improving the epochs from 3 to above 20, both training and test accuracy reached nearly 99% (see Figure 1). This raised doubts about data leakage or other issues. After exploring the data, padding was considered as a possible explanation, since many sequences contain post-padding. However, padding alone did not explain the extremely high performance.

The real issue appeared to be the train-test split. The same users were included in both training and validation sets. This means the model could learn user-specific patterns instead of general gesture patterns. For a realistic scenario, where the system should work for new users, this is not valid. Therefore, a leave-one-user-out strategy was implemented where user U01 was used for the validation (or test) set. I would like to give credit for this solution to my classmate, Steven Bontius.

With this new split, performance became more realistic. The model showed a clear gap between training and test performance and had more difficulty generalizing to unseen users. This confirmed that the initial results were probably unrealistic.

Several architectural experiments were then performed. Replacing the GRU with an LSTM reduced performance. The LSTM also started overfitting earlier. Adding an extra layer and increasing dropout improved the LSTM slightly, but the GRU remained more stable and achieved better overall accuracy.

Next, a 1D convolution layer was added before the GRU. This improved performance, possibly by learning small patterns in the data over time. Increasing the kernel size also led to a small improvement. Finally, enabling pooling resulted in the best overall model. Pooling likely reduced noise and helped the model focus on more important features.

Finally, the best-performing architecture was retrained under different leave-one-user-out splits, using U03 and U06 as validation users in separate runs (see Figure 2).

Performance differed strongly between the two runs. When U03 was used as the validation user, performance was relatively low, while the model performed much better when U06 was used as the validation set. This suggests that U06's gesture patterns are closer to the average patterns learned during training, whereas U03 deviates more from this average. The model is therefore able to capture general gesture behavior, but is not fully generalizable across users. Users who differ more from the overall pattern are harder for the model to generalize to. This remains an important limitation and should be addressed if this was a real-world case.

Loss/test vs Loss/train vs metric/Accuracy ×

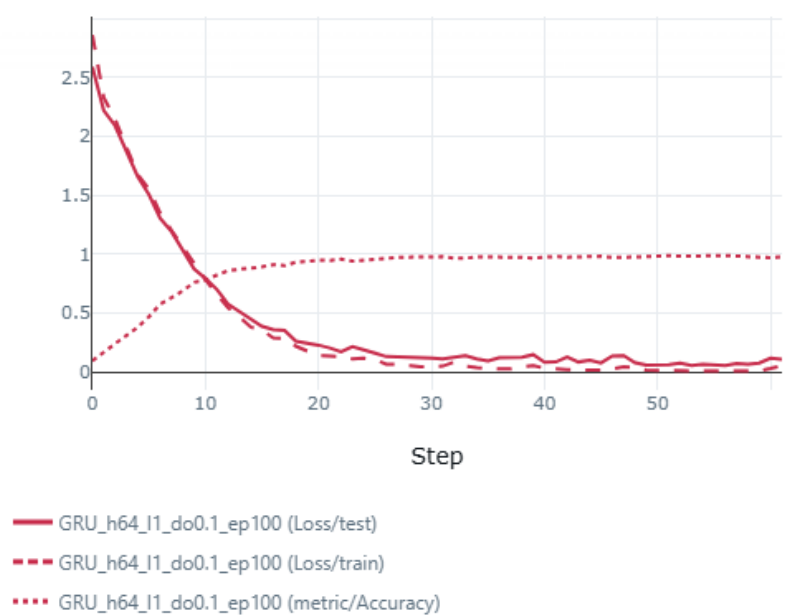


Figure 1 Results after only increasing the number of epochs of the default configuration. Too good to be true?

metric/Accuracy ×

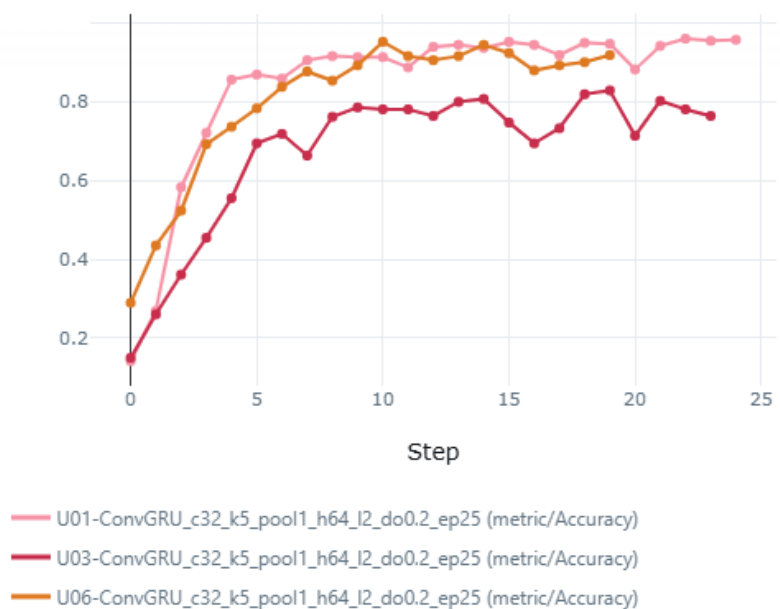


Figure 2 Leave-one-user-out validation results for users U01, U03 and U06